

Anomaly- and Tadpole-Compatible Fermion Completion of Six-Dimensional $SU(4)$ Gauge–Higgs Unification

An exact consistency analysis — machine-checked and independently verified — and the boundary-mass filter

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Abstract

In gauge–Higgs unification the cancellation of gauge anomalies is necessary, but is it sufficient for a viable fermion sector? On an orbifold the answer involves genuinely *local* data — anomalies and tadpoles bound to the fixed points — that a bulk count misses. We make the question concrete for a colored quark block embedded in the dimension-60 representation $(3, \mathbf{60})$ of $SU(4)$ on $T^2/\mathbb{Z}_2$, built over the two-Higgs-doublet gauge sector of Akamatsu, Hirose, Maru and Nago (AHMN) [19]; the fermion embedding is our own. We fix the unique unbroken hypercharge that reproduces the Standard-Model quark charges, and organize every consistency requirement into a reproducible hierarchy of exact conditions: the irreducible six-dimensional and four-dimensional zero-mode anomalies, the localized non-Abelian cubic residues (certified axiom-free in the Lean 4 proof assistant), the localized tadpole, the global $SU(2)$ (Witten) constraint, and the boundary-mass matching of exotic zero modes. We then exhibit — and independently verify in exact rational arithmetic — a 45-multiplet completion that cancels the eight anomaly coefficients and the localized tadpole together, with integral local cubic residues, so that here anomaly freedom and the tadpole condition are *compatible*. This is not accidental. That the two constraints are *independent* was observed long ago by Scrucra, Serone, Silvestrini and Wulzer [15], who left the construction of an anomaly-free spectrum with globally vanishing tadpole as an open problem; we make the independence exact and, beyond it, prove that anomaly-neutral matter realizes *any* tadpole, so *every* anomaly-free completion of the block is tadpole-compatible — settling that open requirement as a structural guarantee, with the exact rank-and-cone test released as a reusable tool. The decisive remaining test, whether the exotics can be given mass without lifting the Standard-Model quarks, is cast as an exact bipartite-matching gate and left open pending the full brane spectrum. Every step is a finite exact computation; the general bounded search is hard and stays open, but the results we report are certificates, not estimates.

1 Gauge–Higgs unification and the subtlety of orbifolds

If a six-dimensional fermion content cancels every gauge anomaly, is the resulting four-dimensional theory automatically healthy? The question is as old as the idea that the Higgs boson need not be an elementary scalar. In 1979 Manton and Fairlie observed that, with extra dimensions, the internal components of a higher-dimensional gauge field transform as four-dimensional scalars, so a Higgs doublet can arise as a low-energy mode of the gauge connection itself [2, 3]. Hosotani

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turned this kinematic remark into a dynamical mechanism: the Wilson-line phases wrapping a compact dimension are physical, and their expectation values break the gauge symmetry [4]. In this *gauge–Higgs unification* (GHU) the Higgs potential vanishes at tree level and is generated radiatively, so the electroweak scale is finite and calculable — the original motivation, and still the appeal: a Higgs mass protected from ultraviolet sensitivity by higher-dimensional gauge invariance.

Two further ingredients make such a picture realistic: chiral fermions, and a breaking of the unified group to the Standard Model. Orbifold compactifications supply both. Kawamura’s construction breaks a grand-unified $SU(5)$ to the Standard Model by boundary conditions on $S^1/(\mathbb{Z}_2 \times \mathbb{Z}'_2)$, projecting out the dangerous colored Higgs while keeping the doublet — and doing so without a symmetry-breaking potential [10]. The economy carries a subtle price. An orbifold is not a manifold; its fixed points are singular loci that host their own, *local*, physics. Anomalies need not cancel point by point in the naive way. Arkani-Hamed, Cohen and Georgi showed that the higher-dimensional anomaly localizes on the fixed planes, so that four-dimensional anomaly cancellation in the low-energy spectrum controls the consistency of the whole construction [11]; von Gersdorff and Quirós developed the localized-anomaly formalism in five and six dimensions and its cancellation by inflow and by brane fields [12, 13]. A second local effect is equally consequential: radiatively generated tadpoles for the internal gauge field (the would-be Higgs) at the fixed points, which feed directly into the Higgs mass and can spoil the calculability that motivated the whole framework. Biggio and Quirós gave the symmetry criteria for when such tadpoles appear, and showed that in six dimensions on $T^2/\mathbb{Z}_2$ they arise for fermions in essentially arbitrary representations [14]. Scrucça, Serone, Silvestrini and Wulzer sharpened the tension into a question. A six-dimensional fermion contributes to the tadpole with a sign fixed by its representation but *independent* of its chirality; so, they observed, the vanishing-tadpole and anomaly-cancellation requirements are “in general independent constraints” — cancelling the anomalies buys nothing for the tadpole, and cancelling the tadpole buys nothing for the anomalies. They then named the missing piece plainly: a realistic model must rest on “an anomaly-free fermion spectrum with a globally vanishing tadpole” [15], and left that as an open problem, with a doubtful note on whether it could be met at all. That question, posed in 2003, is the one this paper takes up. A bulk anomaly count, in short, is necessary but not sufficient — which is the question above, made concrete.

Our arena is the recent six-dimensional $SU(4)$ model of AHMN [19], in which *two* Higgs doublets appear as zero modes of the extra-dimensional gauge field on $T^2/\mathbb{Z}_2$. AHMN specify the gauge and Higgs sectors; the fermion content that would reproduce the Standard-Model quarks is not part of their construction, and it is there that we work. We embed a colored quark block in the dimension-60 representation $(3, \mathbf{60})$ of $SU(4)$ — *our* construction, not theirs — and ask whether it can be completed into a consistent, and ultimately viable, fermion sector. Answering that means passing the full hierarchy of conditions above; our contribution is to impose every one of them *exactly*, as a machine-checkable certificate rather than a case-by-case argument, and to report honestly where the concrete model stands.

Scope and relation to prior work. The general ingredients are established. The mechanism of gauge–Higgs unification in orbifold models, and systematic classifications of anomaly-free orbifold-GUT and GHU matter, are well developed [15, 16, 17, 18]; localized anomalies [11, 12, 13] and localized tadpoles [14] are textbook consistency conditions. We claim no new general physics. What is new here is ours on three counts: (i) the specific $SU(4)/T^2\mathbb{Z}_2$ crossing — integer fermion completion subject to localized anomalies, tadpoles and exotic matching, all at once; (ii) the *exact, reproducible* protocol in which every condition is recomputed exactly and independently verified, with the localized-anomaly fragment machine-checked in Lean 4; and (iii) the structural result that this study yields — our Existence theorem (§9, Theorem 1): that the localized tadpole *never obstructs* an anomaly-free completion, which settles as a structural guarantee the anomaly-free-spectrum-with-vanishing-tadpole problem left open by Scrucça et al. [15]. That the

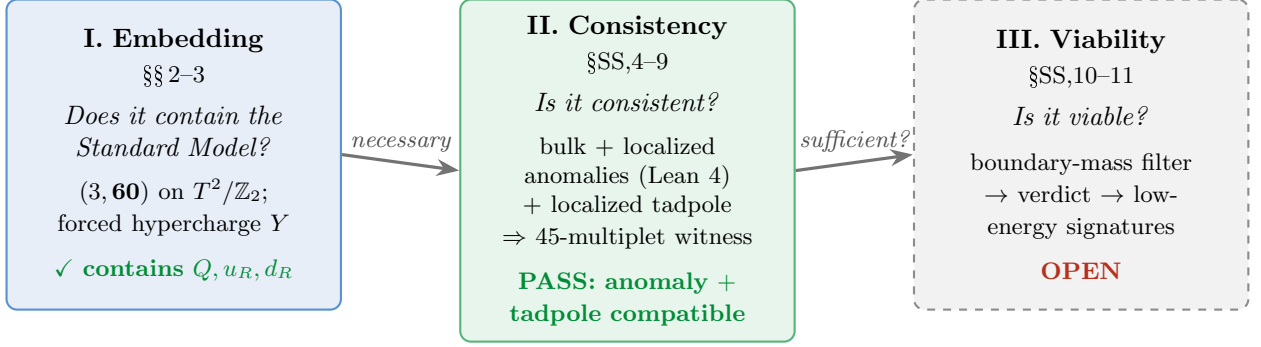


Figure 1: The journey of this paper. We ask three questions in turn — does the representation contain the Standard-Model quarks, is the resulting theory consistent, and is it physically viable — each a stricter test than the last. The embedding and the consistency conditions are settled exactly (the localized anomalies machine-checked in Lean); the viability verdict, gated by the boundary-mass filter, is left open.

Treatment of the fermion sector	Bulk	Localized	Tadpole	Exact	Lean
AHMN gauge–Higgs construction [19]	✓	×	×	×	×
Orbifold-GUT/GHU classifications [16, 17, 18]	✓	✓	×	×	×
This work	✓	✓	✓	✓	✓

Table 1: What each analysis establishes for the fermion sector of this class of six-dimensional orbifold models. AHMN [19] supply the gauge and two-Higgs sector we build on; systematic classifications treat bulk and localized anomalies; the present work adds the localized tadpole, an exact integer completion, and a Lean-checked fragment, imposing all of them together. Checkmarks indicate that the condition is treated for a concrete quark completion, not merely in principle.

two constraints are *independent* was already their observation; our new step is the reachability result that turns independence into a no-obstruction guarantee — made exact, machine-checked, and packaged as a reusable tool. The linear-algebra ingredients (rank tests, Farkas duality) are standard; what is ours is the reachability theorem, its certificates, and their application here. The physical verdict on viability is not yet closed, and we say so throughout.

2 Gauge–Higgs unification on $T^2/\mathbb{Z}_2$

We work on $M^4 \times T^2/\mathbb{Z}_2$. The torus is coordinatized by (y_5, y_6) with identifications $y_i \sim y_i + 2\pi R_i$; the $\mathbb{Z}_2$ acts by $(y_5, y_6) \rightarrow (-y_5, -y_6)$ and leaves four fixed points. The six-dimensional gauge field decomposes into a four-dimensional vector A_μ and two internal scalars A_5, A_6 ; under the orbifold projection the surviving zero modes of $A_{5,6}$ are the Higgs fields. In the AHMN $SU(4)$ model the projection at the four fixed points is governed by two commuting parities,

$$P_0 = P_1 = \text{diag}(+, +, +, -), \quad P_2 = P_3 = \text{diag}(+, +, -, -), \quad (1)$$

so the four fixed points fall into two classes with distinct unbroken local groups,

$$f = 0, 1 : SU(3)_C \times SU(3)_{123} \times U(1)_{15}, \quad f = 2, 3 : SU(3)_C \times SU(2)_{12} \times SU(2)_{34} \times U(1)_P. \quad (2)$$

The four-dimensional gauge group left unbroken everywhere is the intersection $SU(3)_C \times SU(2)_L \times U(1)_Y$ (times a residual $U(1)$), and two Higgs doublets arise from the A_5, A_6 zero modes [19]. The tree-level Higgs potential is the internal field strength, $V_{\text{tree}} \propto \text{Tr}[A_5, A_6]^2$, and vanishes for flat (commuting) configurations. This makes the Higgs elusive by construction: it is a Wilson line, a holonomy phase detectable only by transporting a field once around the compact dimension, so in what sense is it a local field with a mass at all? The answer is the mechanism’s whole point. The physical potential is generated radiatively, of Coleman–Weinberg type [5] in the Wilson-line phase;

because the would-be Higgs is a component of a gauge field, higher-dimensional gauge invariance forbids a local mass counterterm and the potential is finite and calculable, the property that made GHU attractive in the first place [4]. The Higgs’s very non-locality is thus its protection — there is no local operator to renormalize its mass — and a field with no tree-level potential nonetheless chooses a vacuum, born entirely of the quantum response of the tower of modes to the Wilson-line phase. It is exactly this calculability that a localized tadpole endangers: a fixed-point term linear in the internal field feeds directly into the Higgs mass and is *not* protected, so demanding its absence is on the same footing as anomaly freedom, not a cosmetic addition.

Throughout we label an $SU(4)$ weight by the Young entries $n_1 \geq n_2 \geq n_3 \geq n_4$ of the corresponding tableau; the useful internal charges are $q_8 = n_1 + n_2 - 2n_3$, $q_{15} = n_1 + n_2 + n_3 - 3n_4$ and $q_P = n_1 + n_2 - n_3 - n_4$, and the two $\mathbb{Z}_2$ parities of a weight are $p_0 = (-1)^{n_4}$ and $p_2 = (-1)^{n_3+n_4}$. These are the data every exact condition below is built from.

3 The $(3, \mathbf{60})$ quark block and its hypercharge

Before any consistency condition one must ask whether the chosen representation even contains the Standard-Model quarks, and under which unbroken hypercharge. We embed the quarks in the colored block $(3, \mathbf{60})$, the color triplet times the dimension-60 irrep $(0, 2, 1)$ of $SU(4)$. It contains Q , u_R , d_R under a hypercharge that is essentially forced: among the internal mixings it is the unique combination preserving $\sin^2 \theta_W = \frac{1}{4}$ at tree level,

$$\boxed{Y = \frac{-7q_8 + 4q_{15}}{18}}, \quad Q_{\text{em}} = T_3 + Y. \quad (3)$$

Equation (3) reproduces the Standard-Model quark charges exactly (Table 2); everything else in the multiplet is an exotic that a consistent theory must ultimately remove. Two features already visible will drive the rest of the analysis. First, Q_L appears with multiplicity two: one combination is the physical doublet, the other an exotic of identical gauge charges — so lifting the exotic doublet without lifting Q_L is a nontrivial, rank-level question. Second, the exotic zero modes are left-handed doublets and a quartet together with right-handed singlets and triplets: their $SU(2)_L$ representations do not match across chirality, so they cannot pair among themselves and require brane partners. *Established here:* the $(3, \mathbf{60})$ does contain the three Standard-Model quark multiplets under a single, essentially forced hypercharge. *Left open:* the remaining exotic zero modes it also carries, which the consistency gates below must remove without touching Q, u_R, d_R .

The choice of $(3, \mathbf{60})$ is one of economy, not taste. Scanning *all* $SU(4)$ irreps of dimension ≤ 63 , only two have $T^2/\mathbb{Z}_2$ zero modes that contain a complete Standard-Model quark generation (Q_L, u_R, d_R as color triplets) under a single unbroken hypercharge: the $\mathbf{60} = (0, 2, 1)$ used here and its conjugate $(1, 2, 0)$, *both* of dimension 60. No smaller colored $SU(4)$ block does. In that precise, checked sense $(3, \mathbf{60})$ is the minimal representation that can host the quarks at all — which is why it, and not some smaller multiplet, is the natural place to test whether a consistent completion exists. *Why* only these representations, and why so few, is turned from this scanned fact into an exact closed criterion in the companion Part II [1].

4 Bulk anomalies and the reducibility of the color quartic

Which part of a gauge anomaly is an absolute obstruction, and which part can instead be shifted into a reducible Green–Schwarz sector? That distinction organizes what a completion must do. The first requirements are the standard ones: the irreducible six-dimensional gauge and gravitational anomalies, and the four-dimensional anomalies of the surviving zero modes. For

Field	$SU(2)_L$	(q_8, q_{15})	Y	chirality	status
Q_L	2	$(-1, -1)$	$1/6$	L	SM target
u_R	1	$(0, 3)$	$2/3$	R	SM target
d_R	1	$(-2, -5)$	$-1/3$	R	SM target
<i>exotic zero modes</i> : LH doublets/quartet, RH singlets/triplets — to be lifted					

Table 2: Standard-Model quark content of $(3, \mathbf{60})$ under the hypercharge (3), verified exactly. Q_L occurs with multiplicity two — one physical, one exotic — a fact that returns in §10.

$(3, \mathbf{60})$ with $\eta = (+, +)$ the eight coefficients are, exactly,

$$\left[SU(3)_C^4, SU(4)^4, \text{grav}_6, SU(3)^3, SU(3)^2 Y, SU(2)_L^2 Y, Y^3, \text{grav}^2 Y \right] = \left[60, -579, 180, -3, \frac{17}{6}, \frac{23}{2}, \frac{2293}{18}, 17 \right], \quad (4)$$

where the hypercharge is the physical one of Eq. (3) throughout (the four Y -dependent entries scale with that normalization). A subtlety controls what a completion must do with these. The six-dimensional anomaly polynomial factorizes as

$$\text{Tr}_{(3, \mathbf{60})} F^4 = -579 \text{Tr}_4 F_4^4 + 30 C^2 + 426 C X + 864 X^2, \quad C = \text{Tr}_3 F_C^2, \quad X = \text{Tr}_4 F_4^2, \quad (5)$$

and only the primitive quartic $SU(4)^4$ coefficient -579 is irreducible; $SU(3)$ has no primitive quartic Casimir, so $\text{Tr}_3 F_C^4 = \frac{1}{2} C^2$ and the “ $SU(3)_C^4$ ” piece is a reducible $(\text{Tr} F_C^2)^2$. The reducible gauge quadratic form $\begin{pmatrix} 30 & 213 \\ 213 & 864 \end{pmatrix}$ has rank 2. This is where the Green–Schwarz mechanism [6] would ordinarily enter: a two-form field B with coupling $B \wedge \text{Tr} F^2$ cancels an anomaly precisely when the anomaly polynomial *factorizes* as a product of two four-forms, i.e. when the quadratic form is rank one. In six dimensions this generalizes to several tensor multiplets, each contributing one rank-one factor [7], so a rank-two form needs at least two tensors and an explicit factorization. In the published gauge/BLKT setup of AHMN there is no such two-form, so the reducible pieces are imposed as fermionic equations; a multi-tensor branch would have to declare its tensor content and exhibit the factorization. Reducibility is a property of the color factor alone: it means the $SU(3)_C^4$ piece *could* be absorbed by a Green–Schwarz two-form, but it leaves the primitive $SU(4)^4$ index -579 untouched. That one has no such escape — there is no tensor coupling that factorizes a genuine quartic Casimir — so it is an absolute obstruction that the fermion content must cancel outright. Either way the safe, model-independent requirement is to cancel all eight coefficients on the added fermions, which is what we impose. *Established*: the eight coefficients are exact, and the color quartic is reducible so it needs no primitive-quartic tensor. *Left open at this stage*: whether any spectrum can zero all eight together with the localized data — the subject of the gates that follow.

5 Localized anomalies certified in Lean

Here a puzzle sharpens: how can a purely topological quantity, blind to the Wilson-line vacuum, nonetheless constrain the very spectrum that does feel it? Beyond the bulk, the fixed points carry irreducible *local* anomalies. At each $f = 0, 1$ point the pure non-Abelian cubic residues of $(3, \mathbf{60})$ are, exactly,

$$\left[SU(3)_C^3, SU(3)_{123}^3 \right]_{f=0,1} = \left[-\frac{3}{2}, -\frac{3}{4} \right]. \quad (6)$$

The mechanism behind “removal” is anomaly inflow: a fixed point behaves like a defect, and the anomaly of the fermions living on it is cancelled by a current flowing in from the bulk, in the manner Callan and Harvey identified for zero modes on strings and domain walls [8]. Following von Gersdorff [13], fractional residues are a diagnostic of an *incomplete* bulk spectrum: only once the irreducible bulk anomaly is cancelled do the local residues become integral, and only then can

#	Condition	Type	Certificate
1	Bulk 6D irreducible $SU(4)^4$, grav_6	equality	exact integer
2	4D zero-mode anomalies ($SU3^3$, $SU3^2Y$, $SU2^2Y$, Y^3 , grav^2Y)	equality	exact rational
3	Localized cubics [$SU3_C^3$, $SU3_{123}^3$] integrality	congruence	Lean 4
4	Localized tadpole $\Theta = (\Theta_{01}, \Theta_{23}) = 0$	equality	exact integer
5	Global $SU(2)$ (Witten) parity	mod 2	exact
6	Boundary-mass matching (protecting Q, u, d)	rank/matching	exact bipartite

Table 3: The exact consistency hierarchy for a fermion completion. Gate 3 is machine-checked in Lean; the rest are exact Sage / OR-Tools computations.

inflow plus brane-localized Weyl fermions remove them. A completion must therefore make these residues integers — a congruence, not an equality. Two remarks fix the physics. First, the pure non-Abelian cubics carry no $U(1)$ charge, so they are independent of the Wilson line (the Higgs vacuum): the brane content they demand is fixed before electroweak breaking, whereas the mixed $C^2U(1)$ channel does move with the Higgs. Second, because these are finite exact statements about rational numbers, we can certify them by machine. The file `R60LocalizedAnomaly.lean` proves, *axiom-free and sorry-free* in the Lean 4 kernel, the exact integer numerators underlying the residues (-6 and 6 , the rationals following by the trivial normalizations $\frac{1}{4}$, $-\frac{1}{8}$), the dimension count, and that the mixed channel depends on the Wilson line while the pure cubics do not. To our knowledge this is the first fragment of an orbifold-GUT consistency analysis checked in a proof assistant. *Established*, and *machine-checked*: the local cubic residues are exactly $[-\frac{3}{2}, -\frac{3}{4}]$ and become integral only for a completed bulk. *Left open*: the specific brane fermions that carry the compensating inflow, which the boundary spectrum fixes.

6 The localized tadpole and the exact hierarchy

The last local condition before matching is the tadpole. A single bulk Weyl fermion induces, at one loop, a localized tadpole for the internal gauge field with group factor $\Theta_{01} \propto \eta_0 \text{Tr}_R(P_0 T_{15})$ at $f = 0, 1$ and $\Theta_{23} \propto \eta_2 \text{Tr}_R(P_2 T_P)$ at $f = 2, 3$ [14]. For the base block $(3, \mathbf{60})$ we find, exactly, $\Theta_{\text{base}} = (126, -36)$: the isolated block fails the tadpole gate, and a consistent completion must cancel it. Crucially the tadpole is independent of the six-dimensional chirality, so a naive fix by adding vector-like pairs *doubles* rather than cancels it; the cancellation is a genuine constraint on chiral content, coupled to the anomalies. *Established*: the isolated block fails the tadpole gate, $\Theta_{\text{base}} = (126, -36) \neq 0$. *The open question* — whether a single completion can cancel it *together with* all the anomalies — is settled affirmatively in §8.

Table 3 collects the full hierarchy. Gates 1–5 are linear or congruence conditions on the added multiplets; gate 6 is a rank/matching condition. Figure 2 shows their logic: the $(3, \mathbf{60})$ passes gates 1–4 exactly (gate 3 in Lean), while the global $SU(2)$ and the boundary-mass matching remain to be applied.

7 The integer search: reduction and hardness

Imposing gates 1–4 together is a coupled integer feasibility problem. The library of candidate additions — $SU(4)$ irreps of dimension ≤ 35 , all colors, four intrinsic parities, both six-dimensional chiralities — contains 312 species; deduplicating on the exact ten-component signature (eight anomalies, two local cubics, two tadpole components) reduces it to 152 distinct signatures without loss, a 51% reduction that is safe for the linear conditions.

Honesty about the computation matters here, and it turns on a distinction the solvers force on us: when an exact search returns **UNKNOWN**, what has actually been *proved*, and what has merely not been *found*? The *bounded* feasibility problem (“ $\leq k$ added multiplets”) is genuinely hard: exact branch-and-bound over the rationals (PPL) and multi-threaded CP-SAT both return

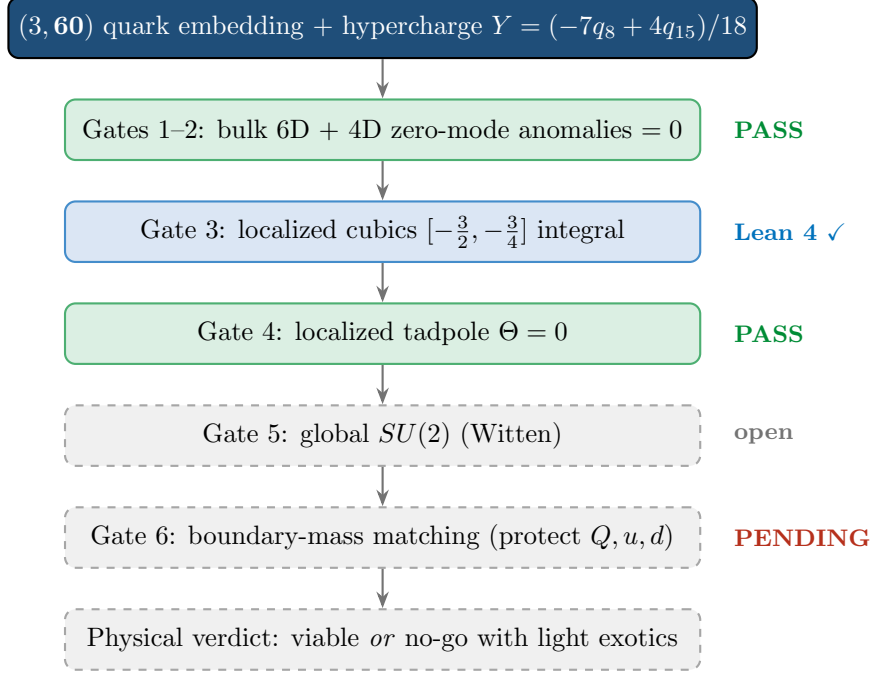


Figure 2: The exact consistency funnel. The 45-multiplet completion of §8 passes gates 1–4 exactly, gate 3 machine-checked in Lean; the global $SU(2)$ and the decisive boundary-mass matching (gate 6) remain. **PASS:** anomalies + tadpole + local integrality; the mass gate pending.

UNKNOWN within budget, and an exact Hilbert-basis attempt (Normaliz) is intractable at this dimension. What we *can* establish is that the obstruction, if any, is not of the cheap kind: a layered certificate shows the linear-programming relaxation is feasible and the integer lattice of the joint (anomaly + tadpole) system is solvable, so no lattice or congruence obstruction rules the completion out. The remaining question is purely one of non-negative integer feasibility — and that we settle constructively in the next section, not by proving a bound. *Established:* no lattice or congruence obstruction rules out a joint completion. *Left open:* the general bounded-search verdict (minimal completion size, or a proof of infeasibility below some k), which we do not claim.

8 An anomaly- and tadpole-compatible completion

An anomaly-free completion of the eight coefficients exists; but the localized tadpole is independent, and it is natural to ask whether both can be met at once. They can. A long-budget exact CP-SAT search on the unbounded joint problem returns an explicit non-negative integer solution, which we then verify independently in exact rational arithmetic.

Proposition 1 (Anomaly- and tadpole-compatible completion). *The (3, 60) block admits a completion by 45 added multiplets that cancels all eight bulk/zero-mode anomaly coefficients and the localized tadpole Θ , with integral localized cubic residues.*

Independent verification in exact rational arithmetic gives

$$\boxed{\sum_{45}(8 \text{ anomalies}) = \mathbf{0}, \quad \sum \Theta = (-126, 36) = -\Theta_{\text{base}}, \quad [SU3_C^3, SU3_{123}^3]_{\text{res}} = [2, 6] \in \mathbb{Z}^2}. \quad (7)$$

We stress the honest scope. This is not a repair of the earlier eight-anomaly witness of 27 multiplets, which has residual tadpole $(126, -20) \neq 0$ and *fails* the tadpole gate; nor is it an optimal or bounded solution — the general bounded problem remains open. It is a constructed, exactly verified spectrum satisfying gates 1–4 jointly. Does exhibiting one such completion make the embedding *special*, or only show it is not immediately dead? Honestly the latter — existence

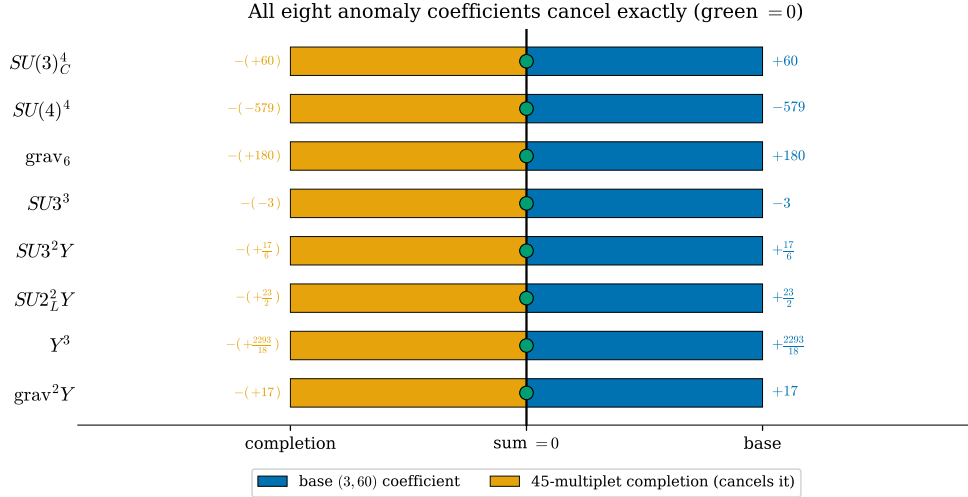


Figure 3: The eight bulk and four-dimensional zero-mode anomaly coefficients of the base $(3, \mathbf{60})$ block (right, blue) and the exactly opposite contribution of the 45-multiplet completion (left, orange); each pair sums to zero (green). Coefficients are exact rationals, recomputed independently.

removes a candidate no-go, it does not prove the completion unique or generic. But that is exactly the physical upshot that matters here: contrary to the naive worry that a nonzero localized tadpole kills every anomaly-free completion, the tadpole condition is *compatible* with anomaly freedom for $(3, \mathbf{60})$.

The spectrum is shown in Figure 5 and listed in full — all 17 species — in Table 4. A single witness, though, still leaves the central question open: is this completion a lucky needle, or the visible face of something inevitable? That is the subject of the next section. *Established:* anomaly freedom and localized-tadpole cancellation are jointly realizable for $(3, \mathbf{60})$ — orbifold-level consistency is not obstructed. *Left open (here):* whether it is *always* so, settled next; and whether the spectrum is physically viable, which only the boundary-mass filter of §10 can decide.

9 From a witness to a guarantee: the tadpole never obstructs

Every model-builder knows the quiet doubt that trails a lucky find: was that a law, or just this once? The 45-multiplet spectrum of the previous section cancels the anomalies and the tadpole together — but one example is a coincidence until shown otherwise, and the whole temper of this paper is to prefer a certificate to a coincidence. So we press the sharper question the witness cannot answer on its own: is joint cancellation *inevitable*, or did $(3, \mathbf{60})$ simply get lucky? The answer, it turns out, needs no further search — only two elementary facts, because the anomalies and the tadpole are both *linear* functionals of the added multiplicities. Write A for the anomaly map (to $\mathbb{Q}^8$) and Θ for the localized-tadpole map (to $\mathbb{Q}^2$) on the library of additions. Two questions then decide everything. First: are the tadpole and the anomalies *independent* dials, or is there a hidden combination $\varphi \circ \Theta$ that stays fixed once the anomalies are cancelled — a conservation law that would pin the tadpole away from zero? Second: since one may only *add* matter, never subtract, can the additions that leave all eight anomalies untouched still push the tadpole in *every* direction? Both answers turn out favorable, and exactly so.

Return now to the question left open in the introduction — the one Scrucca, Serone, Silvestrini and Wulzer posed in 2003. For two decades it stood dry: everyone agreed that a realistic model needed an anomaly-free spectrum with a globally vanishing tadpole, everyone knew the two constraints were independent, and yet the joint solution was treated as a model-by-model hope — something to stumble on, not something guaranteed. The two questions above end that drought. The first, independence, was already their observation; our part is to make it *exact* — a full-rank

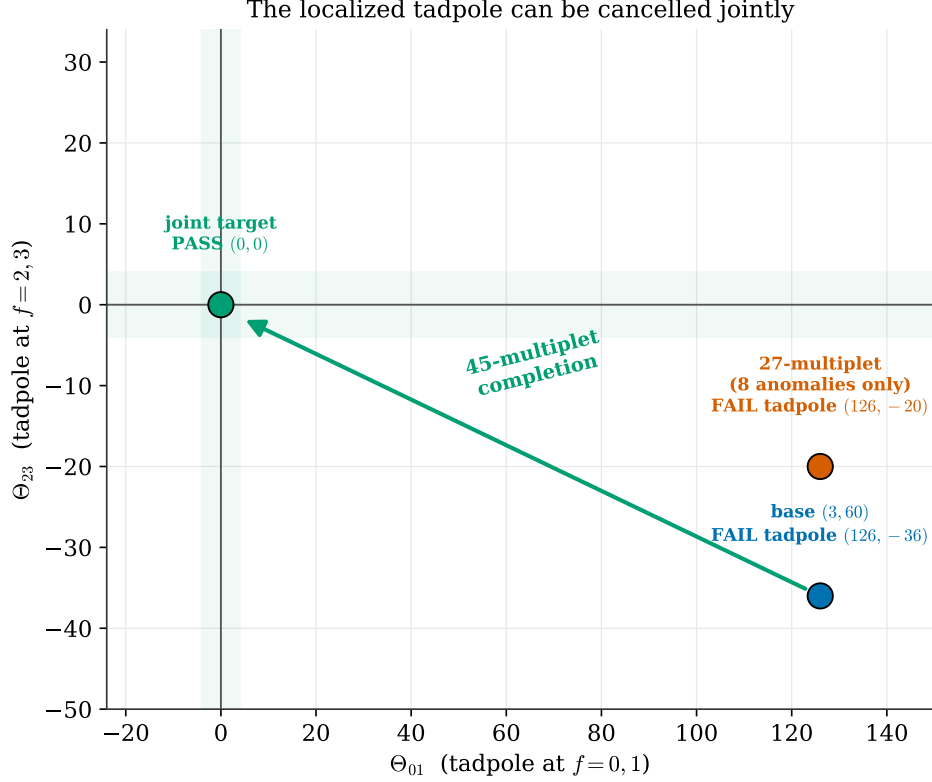


Figure 4: The localized-tadpole plane $(\Theta_{01}, \Theta_{23})$. The base block sits at $\Theta_{\text{base}} = (126, -36)$ and *fails* the tadpole; the eight-anomaly witness of 27 multiplets moves only to $(126, -20)$ and still fails; the 45-multiplet completion (green arrow) brings the total to the origin. Anomaly freedom alone does not cancel the tadpole — but a joint completion does.

determinant certificate over the entire library, not a remark in passing. The second question is the genuinely new one, and its answer is the turn of the story: anomaly-neutral matter reaches *every* tadpole value. That single fact converts independence into a no-obstruction *guarantee*, and settles the twenty-two-year-old requirement — not for one fortunate spectrum, but structurally, for the whole class at once. It is our main structural result.

Why did the question stand for so long? Not for any want of effort — the field circled it for two decades — but because it was being asked in the wrong language, and sought in the wrong place. Model-building, by long habit, is a search: choose a fermion content, ask whether the anomalies cancel *and* the tadpole vanishes, and when they do not — as, almost always, they do not — choose another and begin again. It is a treadmill, and every failed step murmurs the same discouraging suggestion, that perhaps no step exists at all; SSSW’s closing note carries exactly that fatigue. And the roads the field *did* take led elsewhere. One tradition sought to *forbid* the tadpole outright, by arranging a symmetry under which it cannot arise — the “Higgs-gauge unification without tadpoles” of Biggio and Quirós [14] — a genuine escape, but one open only in special configurations, and no help at all in cancelling a tadpole that is already there. Another leaned on the oldest reflex for anomalies: add *vector-like* matter, which kills them for free. Yet the localized tadpole is blind to chirality, so a vector-like pair *doubles* it rather than relieving it — the most familiar instrument in the model-builder’s hand is, here, exactly the wrong one. What the problem quietly asked for was the opposite gesture: to add *chiral* matter, each species anomalous on its own yet summing to nothing, and enough of it to steer the tadpole home — an unnatural thing to attempt by hand, and an unnatural thing for a minimalist even to want. Independence, the very thing SSSW had noticed, is moreover a false dawn here: knowing that the two dials turn separately does not promise that both can be set to zero at once, because the model-builder is

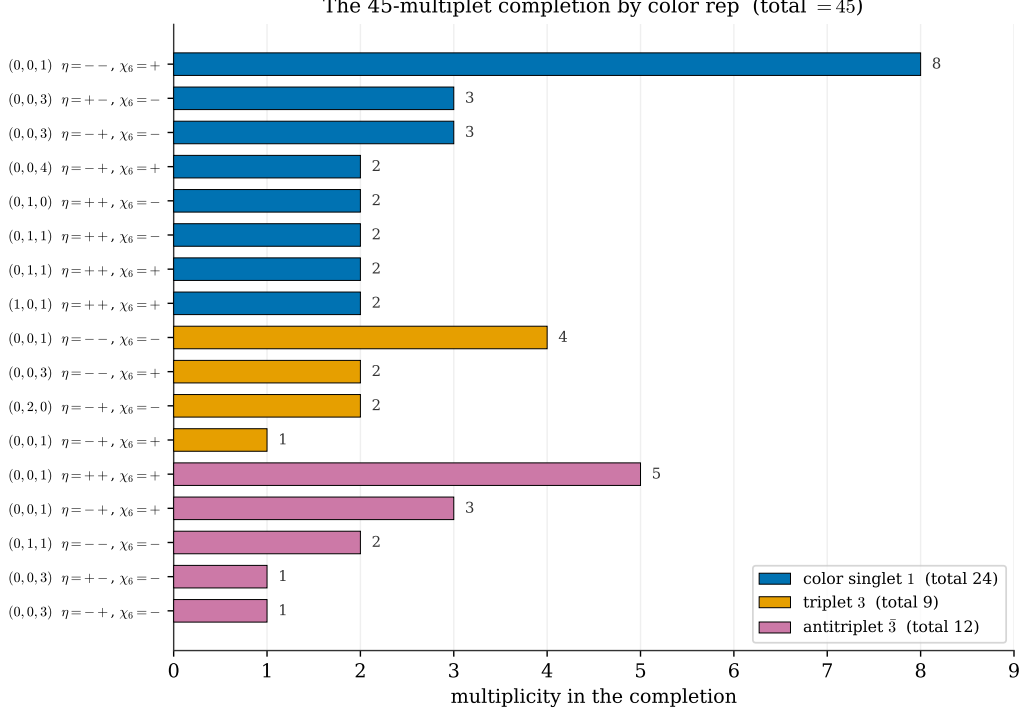


Figure 5: The 45-multiplet completion, grouped by color representation (Dynkin label, intrinsic parity η , six-dimensional chirality χ_6 ; multiplicities sum to 45). **Verified: eight anomalies + localized tadpole + local cubic integrality; the mass gate (§10) pending.**

permitted only to *add* matter, never to take it away. The true object is therefore not a pair of independent numbers but a *cone* — the set of tadpoles reachable by piling on anomaly-neutral matter — and a cone can be lopsided, opening toward one horizon and walled off toward another. No one had drawn that cone. Two instincts kept it in the dark. The first is a matter of vocabulary: the existence of *some* completion, among all of them, is a feasibility question — of ranks, integer lattices, Farkas certificates — and those are the instruments of combinatorial optimization, a neighbouring country the model-builder seldom visits, so the question was rarely even put in a form that could be settled once and for all. The second is a matter of taste, and the more human of the two: the completion that works is not lean but lavish — forty-five multiplets poured in — and a discipline raised to admire economy does not go digging where the answer is heavy. The solution sat in plain sight, in the one direction that aesthetics had told everyone not to look. Trade the search for a feasibility test, set aside the reflex for minimality, carry in the optimizer’s tools — and the twenty-two-year wall thins to nothing: a rank, and a cone.

Theorem 1 (Existence). *Over the $SU(4)$ library (dimension ≤ 35 , all colors, parities and six-dimensional chiralities), the combined map (A, Θ) has full rank $10 = 8 + 2$, and the image of Θ on the anomaly-neutral non-negative cone $\{x \geq 0 : Ax = 0\}$ is all of $\mathbb{R}^2$. Consequently the localized tadpole imposes no obstruction beyond anomaly freedom: every anomaly-free completion of $(3, \mathbf{60})$ admits a tadpole-compatible one. The 45-multiplet spectrum is one realization, representative rather than exceptional.*

The proof is two exact certificates. *Independence:* the stacked map $[A; \Theta]$ has rank 10 — an explicit 10×10 integer minor has determinant $3.6 \times 10^{20} \neq 0$ — so the two tadpole components are not any combination of the eight anomaly coefficients and no invariant $\varphi \circ \Theta$ is conserved. (Were the rank deficient, such a conserved combination would exist; the earlier anomaly-free 27-multiplet spectrum, which still carries tadpole $(126, -20) \neq 0$, already shows the tadpole is genuinely not fixed by the anomalies.) *Reachability:* anomaly-neutral non-negative additions

#	Dynkin (a, b, c)	color	$\eta = (p_0, p_2)$	χ_6	mult.
1	(0, 0, 1)	1	(−, −)	+	8
2	(0, 0, 3)	1	(+, −)	−	3
3	(0, 0, 3)	1	(−, +)	−	3
4	(0, 0, 4)	1	(−, +)	+	2
5	(0, 1, 0)	1	(+, +)	−	2
6	(0, 1, 1)	1	(+, +)	−	2
7	(0, 1, 1)	1	(+, +)	+	2
8	(1, 0, 1)	1	(+, +)	+	2
<i>color singlets subtotal</i>					24
9	(0, 0, 1)	3	(−, +)	+	1
10	(0, 0, 1)	3	(−, −)	−	4
11	(0, 0, 3)	3	(−, −)	+	2
12	(0, 2, 0)	3	(−, +)	−	2
<i>triplets subtotal</i>					9
13	(0, 0, 1)	$\bar{\mathbf{3}}$	(+, +)	+	5
14	(0, 0, 1)	$\bar{\mathbf{3}}$	(−, +)	+	3
15	(0, 0, 3)	$\bar{\mathbf{3}}$	(+, −)	−	1
16	(0, 0, 3)	$\bar{\mathbf{3}}$	(−, +)	−	1
17	(0, 1, 1)	$\bar{\mathbf{3}}$	(−, −)	−	2
<i>antitriplets subtotal</i>					12
total					45

Table 4: The complete 45-multiplet completion: all 17 species by $SU(4)$ Dynkin label (a, b, c) , color representation, intrinsic parity $\eta = (p_0, p_2)$, six-dimensional chirality χ_6 , and multiplicity. This is the spectrum that cancels the eight anomaly coefficients and the localized tadpole with integral local cubic residues (§8); independently re-verified in exact rational arithmetic by `r60_verify_joint.sage`.

realize each signed tadpole axis — $\Theta = (288, 0)$ by two multiplets, $(0, 48)$ by four, and their negatives likewise. Four directions that positively span the plane put 0 in the interior of the achievable cone, so whatever residual tadpole an anomaly-free embedding carries, anomaly-neutral matter cancels it (Figure 6). Everything is exact rational linear algebra, and the four witnesses together with their independence are re-checked in the Lean 4 kernel (`R60JointExistence.lean`, depending only on `propext`). $\square$

Behind the linear algebra is a picture worth keeping. The tadpole is a *soft* quantity — a signed, parity-weighted sum whose sign flips with the intrinsic parity and six-dimensional chirality of the matter one adds — while the anomaly polynomials are rigid. The fermion catalogue is thus rich enough to *steer* the tadpole wherever one likes, in any direction, without nudging a single anomaly coefficient. Read this way the 45-multiplet spectrum was never a needle in a haystack: the haystack is made of needles, and we merely picked one up.

And the method outlives the example. The argument’s form is model-independent: for *any* orbifold fermion library the same **rank + cone** test decides whether a localized functional is an independent obstruction (a conserved invariant, or a pointed cone — an immediate no-go when the base violates it) or is always absorbable. We release it as the ancillary tool `ghu_preflight.py`: given a fermion library (representations, colors, parities, chiralities) and a base block, it returns the exact verdict — UNOBSTRUCTED, NO-GO or CONE-OBSTRUCTED — with its rank/Farkas certificate, the physics model a single pluggable callback. Per model one obtains a decisive verdict; when obstructed, the explicit conserved invariant $\varphi \circ \Theta$ (a candidate low-energy signature); when unobstructed, a guarantee that an integer search terminates successfully together with the constructive Farkas directions to *build* a completion without a solver; and, run over a family of embeddings, a classification of the landscape into obstructed and unobstructed. It is thus an exact preflight before an expensive search — decision-complete on both sides — and a way to hunt the genuinely obstructed models, where anomaly freedom fails to imply tadpole freedom.

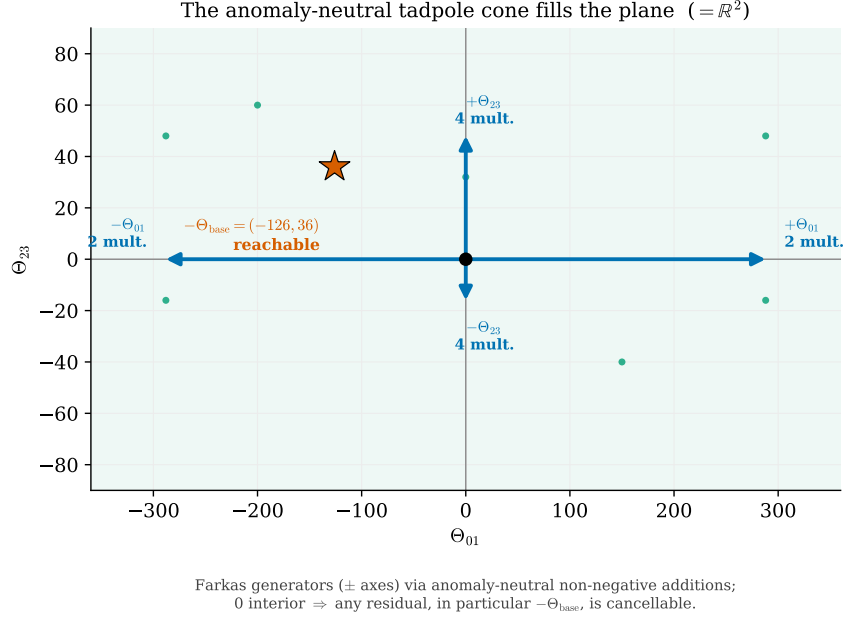


Figure 6: Theorem 1, visually. Anomaly-neutral non-negative additions realize both signs of each tadpole axis (blue arrows, labeled by the number of multiplets — the Farkas certificates), so the cone they generate is the whole plane $\mathbb{R}^2$ (shaded). The origin is interior; hence the base residual $-\Theta_{\text{base}} = (-126, 36)$ (star) — and any other — is cancellable by adding anomaly-neutral matter.

With that, the middle act of the story closes, and cleanly: the $(3, \mathbf{60})$ block is *consistent*, and provably so — by a law, not an example. *Established:* tadpole cancellation is no obstruction beyond anomaly freedom; every anomaly-free completion admits one. *Left open:* the fully general integer statement over all embeddings (a finite Hilbert-basis check — the rational cone is already the whole plane). What the block is not yet known to be is *viable*; and for that the last and strictest question of the journey remains.

10 The boundary-mass filter and the physical verdict

A completion that is anomaly- and tadpole-consistent still carries exotic zero modes. Boundary masses can pair them off, but the sharp question is whether they can do so while leaving Q , u_R , d_R massless. A localized Dirac mass at a fixed point pairs a bulk zero mode with a brane Weyl fermion of conjugate local representation and opposite four-dimensional chirality; whether *all* exotics can be so lifted while every target stays in the kernel is exactly a bipartite-matching (generic-rank) condition on the allowed-mass graph. This is a finite, exact test — it terminates with a definite verdict rather than a timeout — and we have validated the solver on known-answer cases. Two structural facts sharpen it: the exotic left- and right-handed zero modes carry mismatched $SU(2)_L$ representations, so brane partners are unavoidable; and the multiplicity-two Q_L means one must lift one degenerate doublet while protecting its twin, a rank-deficiency that is possible but not automatic.

Running this gate against the full brane library at the two fixed-point classes, and imposing the global $SU(2)$ parity that forbids an odd number of Weyl doublets (the Witten anomaly [9]), is the decisive physical step. We report it as open: the verdict determines whether the $(3, \mathbf{60})$ completion is viable — a realistic quark sector of the AHMN two-Higgs-doublet model — or a no-go, in which case the obstruction carries a concrete low-energy signature (unavoidable light exotics or a vector-like partner of a Standard-Model quark). Either outcome is a result of the exact hierarchy; neither is claimed here.

11 Low-energy signatures of the two verdicts

Grant the exact hierarchy its two possible answers — what would each look like in an experiment? The value of pinning the consistency question down is that the two outcomes are not phenomenologically interchangeable: each maps to a distinct, and in principle falsifiable, low-energy signature. One robust prediction survives either way. Because the underlying AHMN construction delivers *two* Higgs doublets as Wilson-line zero modes [19], the scalar sector is that of a two-Higgs-doublet model, with the attendant extra states H , A , $H^\pm$ and the flavor-alignment questions such models carry [20]; in gauge–Higgs unification their potential is radiative and finite, so the doublet masses and mixings are tied to the compactification dynamics rather than free [4]. Our analysis does not compute those masses, but it fixes the setting in which they live.

If the boundary-mass filter *passes* — every exotic lifted at the compactification scale $1/R$ while Q, u_R, d_R stay massless — the light theory is the Standard Model plus that second doublet, and the distinctive high-scale imprint is a tower of Kaluza–Klein excitations of the gauge and matter fields at $1/R$, the generic fingerprint of the extra dimensions. If instead the filter returns a *no-go*, the obstruction is not abstract: the rank-deficiency of the matching leaves a physical remnant. Two shapes are possible. Either an exotic zero mode cannot be paired at all and survives as an unavoidable light state — a colored or weak-charged particle near the electroweak-to-TeV window, the sharpest possible falsifier — or the only way to lift the exotic doublet is to pair it with a partner that also marries a Standard-Model quark, leaving a *vector-like quark*: a non-chiral partner of Q , u_R or d_R that mixes with the light generations and is searched for directly at colliders through single and pair production [21]. The multiplicity-two Q_L isolated in §3 is exactly what makes this second option natural — two doublets of identical charges, one of which must be given a vector-like mass — so a vector-like Q -partner is the specific signature a no-go here would carry.

How heavy, and how would one tell them apart? Gauge–Higgs unification ties the compactification scale $1/R$ to the radiatively generated Higgs mass; reproducing $m_h = 125$ GeV with a top-driven Coleman–Weinberg potential typically pushes $1/R$ into the multi-TeV range, so the Kaluza–Klein tower and the heavier scalars of the two-doublet sector would sit there, at or beyond current collider reach. A vector-like quark need not, however, live at $1/R$: if it arises from a boundary-mass rank-deficiency, its mass is a brane parameter and could be considerably lighter, within the reach of the LHC and its high-luminosity phase. Such a state would announce itself in the established vector-like-quark channels [21] — pair production followed by decays to Wq , Zq or hq , or single production through its mixing with the light generations — and the multiplicity-two Q_L points specifically to a partner in the (t, b) doublet sector. The other no-go branch, an exotic that cannot be paired at all, would instead surface as a stable or long-lived colored or weak-charged state near the electroweak scale, a sharper and more immediately falsifiable signal. The distinction is therefore not academic: one verdict places the new physics at $1/R$ far above, the other a vector-like quark possibly already in range. None of these masses is computed here — they are downstream of the mass filter and the compactification data — but the *classes* of signal are fixed by the structure established above.

Established: whichever way the verdict falls, the framework robustly predicts a second Higgs doublet, and the two verdicts separate cleanly into a Kaluza–Klein/2HDM spectrum on the one hand and a light-exotic or vector-like-quark signature on the other. *Left open*: which of these the model realizes, and the scale $1/R$ that sets the masses — neither computed here, both downstream of the mass filter.

12 Consistency, viability, and what an exact answer buys us

What does this concrete exercise teach about the gap between a consistent theory and a viable one? The case sharpens a distinction easy to blur. *Consistency* — anomaly freedom, tadpole cancellation — is a statement about the ultraviolet; *viability* — a light spectrum that is the

Standard Model and nothing else — is a statement about the infrared. Gauge–Higgs unification makes the seam visible: the Higgs is a Wilson line, so the same orbifold data that fix the anomalies also shape the Higgs potential, yet the anomaly obstruction is topological and blind to the Higgs vacuum, while the mass of the exotics is a spectral, infrared question. A model can be perfectly consistent and still fail to be viable, and only an infrared filter — here the boundary-mass matching — decides.

There is a reflection about the Higgs itself, and why it stays so hard to pin down. In gauge–Higgs unification the same internal components A_5, A_6 are simultaneously gauge field and Higgs; which of the two the theory shows depends on the vacuum it sits in, so the Higgs has no fixed identity independent of the Wilson line it rides. That elusiveness is not a nuisance but the source of its good behavior — an object with no local definition admits no local mass counterterm — and it is exactly what the localized tadpole threatens to spoil, by feeding a fixed-point term straight into a mass the mechanism was built to protect. Read this way the entire consistency hierarchy is a set of conditions for keeping the Higgs elusive: anomaly freedom and tadpole cancellation are the price of a scalar that is protected precisely because it is never quite a scalar. The completion we exhibit pays that price at the orbifold level; whether the infrared then delivers a Standard-Model Higgs and nothing else is the question the mass filter holds.

There is a methodological reflection too. Where, when the arithmetic is machine-checked, does the last freedom to be wrong live? Not in the numbers — in the sentence that names them. When each link of the argument is a machine-verifiable certificate, the failure mode of model-building shifts: the danger is no longer an arithmetic slip but a *misinterpretation* — a true number attached to a false sentence. Making the pipeline exact does not remove that danger, but it isolates it, and lets a wrong turn be caught by the ground truth rather than by taste. Several plausible intermediate claims in this very study were refuted that way before entering the paper — a factor-of-two normalization in the hypercharge, and a proposed vector-like correction of the tadpole that the integer lattice ruled out. In that sense the contribution is as much about *how* such questions can be settled as about the particular answer for (3, 60).

13 Reflections on the problem, and the directions it opens

Step back from (3, 60) for a moment. In orbifold model-building the local data — localized anomalies, tadpoles, boundary masses — have long been treated as a checklist of separate hazards, each cleared by hand. What the exact view adds is that these conditions have an *algebra*: some are rigid and mutually constraining (the anomaly polynomials), others soft and freely steerable (the tadpole), and whether a given local condition is a genuine obstruction is not a matter of luck to be discovered by search but a structural fact to be computed once. A landscape of case-by-case checks becomes a small number of decidable questions. Three threads follow naturally, and we flag them as proposals rather than promises.

First, close the concrete case. The one verdict still open for (3, 60) is viability: assembling the brane library at the two fixed-point classes and running the boundary-mass matching of §10 is a finite, exact test with a definite yes/no — either a realistic quark sector of the AHMN two-doublet model, or a certified no-go with a concrete signature (a light exotic or a vector-like partner).

Second, harden and machine-check the structure. The Existence theorem is proved over $\mathbb{Q}$ with the integer statement settled for (3, 60) by the witness; the fully general integer form is a finite Hilbert-basis step. Both it and the rank/Farkas certificates are of a shape a proof assistant handles well, extending the localized-anomaly Lean fragment to a verified *structural* result — a rare thing for a model-building theorem.

Third, and most invitingly, sweep the landscape for obstructions. The same **rank + cone** test — packaged as `ghu_preflight.py` — applies to any orbifold group and matter ($SU(5)$, $SU(6)$, E_6 , other representations) and to other fixed-point data (localized Fayet–Iliopoulos terms, further local functionals). The valuable target is not to confirm more unobstructed cases but to *hunt the obstructed ones*: embeddings where anomaly freedom demonstrably fails to force tadpole freedom.

Each such case carries a conserved invariant $\varphi \circ \Theta$ that is a concrete low-energy signature — a principled search for genuine no-go models, rather than one construction at a time. In the larger picture this is a small instance of a program worth pursuing wherever a model-building result currently reads “we found one”: to ask whether the find was a law, and to turn the computational witness into a structural theorem. The companion Part II [1] carries out exactly this program for the minimality found above, turning the scanned fact that $(3, \mathbf{60})$ is the smallest admissible representation into an exact classification of *which* $SU(4)$ representations contain the Standard-Model quark cell.

14 Methodology and reproducibility

Every claim above is a finite exact computation. Anomaly and tadpole coefficients are exact rational sums over $SU(4)$ weights (SageMath); the coupled feasibility is studied with exact branch-and-bound (PPL), OR-Tools CP-SAT, and layered lattice certificates (Smith normal form), with the 45-multiplet witness re-verified in exact rationals; the boundary-mass gate is exact bipartite matching; and the localized-cubic fragment is certified axiom-free in Lean 4. We are explicit about what is *not* established: the general bounded search is UNKNOWN, and a Hilbert-basis (Normaliz) certificate was not produced at this dimension. The ancillary material contains the scripts, the full 45-multiplet spectrum, the constraint table with each condition’s status and scope, and the Lean source, so that every number here can be regenerated. The exact computations were carried out and cross-checked by two independent agents working against this common ground truth; that working method, rather than any single number, is what most reduced the interpretation errors a model-building analysis is prone to.

Table 5 lists the machine-checkable artifacts released with this paper; each entry regenerates a specific claim above.

Artifact	Certifies	Tool
R60LocalizedAnomaly.lean	localized cubic residues, $\dim = 60$, Wilson dependence	Lean 4 (axiom-free)
R60JointExistence.lean	the Existence theorem’s cone/independence certificate	Lean 4 (only prope xt)
r60_sm_embedding.sage	hypercharge $Y = (-7q_8 + 4q_{15})/18$; SM charges	SageMath (exact)
r60_anomalies.sage	eight bulk/zero-mode coefficients of $(3, \mathbf{60})$	SageMath (exact)
r60_tadpole.sage	localized tadpole $\Theta_{\text{base}} = (126, -36)$	SageMath (exact)
r60_cpsat.sage	the 45-multiplet joint completion search	OR-Tools CP-SAT
r60_verify_joint.sage	independent re-check of the 45-multiplet witness	SageMath (exact)
r60_n7_matching.sage	boundary-mass bipartite gate (validated cases)	OR-Tools / exact
structural_theorem.sage	$\text{rank}[A; \Theta] = 10$ and the tadpole cone (Thm 1)	SageMath (exact)
ghu_preflight.py	reusable rank+cone obstruction oracle (any model)	SageMath (exact)
make_paper_graphics.py	Figures 3–5 from the verified data	Matplotlib

Table 5: Reproducibility manifest. Each artifact regenerates one claim in the paper; the Lean file is checked by the kernel with no axioms beyond Lean’s own and no **sorry**. Released with the ancillary material.

15 Conclusion

The main lesson is concrete, and it closes a question older than the model at hand. Anomaly freedom and localized-tadpole cancellation — named *independent* constraints by Scrucra, Serone, Silvestrini and Wulzer in 2003, who left the construction of an anomaly-free spectrum with a globally vanishing tadpole as an open problem — are here not merely compatible: the tadpole is proved to *never* obstruct an anomaly-free completion, so that after two decades their requirement is met not by luck but as a structural guarantee, exact and machine-checked. The $(3, \mathbf{60})$ block, which a scan shows to be the minimal representation able to host the quarks at all, realizes it explicitly — a 45-multiplet completion cancelling all eight anomaly coefficients and the localized

tadpole *together*, with integral local cubic residues and two Lean-checked certificates. What remains is the sharpest question of all — whether that consistent spectrum is also physically viable, its exotics lifted and only the Standard Model left light — and it is the boundary-mass filter, itself an exact and finite test, that will decide it. Consistency, here, has been settled with certificates; viability is the next certificate to earn.

A Group-theoretic conventions and the exact indices

So that every number above can be regenerated, we fix here the conventions the Lean and Sage artifacts encode. An $SU(4)$ weight is written by its Young entries $n_1 \geq n_2 \geq n_3 \geq n_4$; the internal charges used throughout are

$$q_8 = n_1 + n_2 - 2n_3, \quad q_{15} = n_1 + n_2 + n_3 - 3n_4, \quad q_P = n_1 + n_2 - n_3 - n_4, \quad (8)$$

and the two orbifold parities of a weight are $p_0 = (-1)^{n_4}$, $p_2 = (-1)^{n_3+n_4}$. The hypercharge (3), $Y = (-7q_8 + 4q_{15})/18$, is the unique internal combination assigning Q_L, u_R, d_R their physical charges $1/6, 2/3, -1/3$ (Table 2) and preserving $\sin^2 \theta_W = \frac{1}{4}$ at tree level; the four Y -dependent anomaly coefficients are quoted in this same normalization.

The four-dimensional zero-mode anomalies use the Dynkin index of the $SU(2)_L$ spin content, $T_2(d) = \frac{1}{12}d(d^2 - 1)$ for an irrep of dimension d , weighted by the color dimension and the six-dimensional chirality sign; the pure cubic $SU(3)^3$ uses the fundamental cubic normalization. The six-dimensional coefficients use the quartic index A_4 of the $SU(4)$ irrep (the coefficient of the primitive $\text{Tr}_4 F^4$ after subtracting the reducible $(\text{Tr} F^2)^2$ part) and the representation dimension. For the $SU(3)$ color factor there is *no* primitive quartic Casimir, so $\text{Tr}_3 F_C^4 = \frac{1}{2}(\text{Tr}_3 F_C^2)^2$ and the “ $SU(3)_C^4$ ” entry is reducible; this is the origin of the rank-two quadratic form of §4. The localized cubic residues at $f = 0, 1$ carry the fixed-point normalizations $\frac{1}{4}$ (color) and $-\frac{1}{8}$ ($SU(3)_{123}$), and the tadpole components are $\Theta_{01} \propto \sum \eta_0 p_0 q_{15}$ and $\Theta_{23} \propto \sum \eta_2 p_2 q_P$ over the weights, times the color dimension.

Under $SU(3)_C \times [SU(3)_{123} \times U(1)_{15}]$ the block (3, **60**) with $\eta = (+, +)$ decomposes into the electroweak zero modes of Table 2 plus the exotics; the 45-multiplet completion is listed by (Dynkin, color, η, χ_6) with multiplicities in full in Table 4 and in the ancillary `r60_verify_joint.sage`. All indices in this appendix are recomputed exactly (no floating point) by that script and by `R60LocalizedAnomaly.lean`.

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